DBMS LAB — EXPERIMENT 04

Joins, Subqueries and EXPLAIN

Note: These are expected/representative outputs based on the described SQL and sample data, not screenshots from a live MySQL run. Exact formatting and EXPLAIN plans can vary by MySQL version.

Output 1: INNER JOIN — employee and department

The join returns 30 rows when every employee has a valid department. Sample:

Emp_ID	Emp_Name	Dept_Name
1	Aarav Sharma	Human Resources
2	Ananya Gupta	Human Resources
6	Rahul Kumar	Information Technology
13	Aman Yadav	Finance
…	…	…

Output 2: LEFT JOIN — all departments and employees

All five departments appear. If a department has no matching employee, its employee columns appear as NULL. With the supplied data, each department has employees.

Dept_ID	Dept_Name	Employee_Count
1	Human Resources	5
2	Information Technology	7
3	Finance	6
4	Marketing	6
5	Research and Development	6

Output 3: SELF JOIN — pairs of employees in the same department

Employee_1	Employee_2	Dept_ID
Aarav Sharma	Ananya Gupta	1
Aarav Sharma	Riya Verma	1
Ananya Gupta	Riya Verma	1
Rahul Kumar	Neha Sharma	2
…	…	…

Output 4: Subqueries / EXISTS / EXPLAIN

Subquery and EXISTS results depend on the exact predicates used in the experiment script. EXPLAIN returns optimizer details such as table, access type, possible keys, selected key, estimated rows, and Extra; its values vary by MySQL version and indexes.

Use these tables as representative output formatting. Run the experiment SQL in MySQL to capture the exact full result sets.